

Consumer Theory for the Badly Behaved

Abstract

We introduce a new method suitable for analyzing consumers who do not satisfy traditional revealed preference axioms. We show that to every dataset of consumer behavior there exists a utility function which not only explains the data but is also better-behaved (in some appropriately defined sense) than every other utility function which also explains the data. We show how these “best-behaved” utility functions can be used to make statements about consumer well-being and to make demand predictions. We also provide a method to quantify the degree to which the consumer fails to act as a well-behaved utility maximizer. The advantage of our approach is (i) it can be used even when the consumer does not act according to revealed preference axioms, (ii) it agrees with the traditional approach to consumer theory when the consumer does satisfy these axioms, and (iii) all of our methods are easy to implement in practice.

1 Model

1.1 Definitions

Let X be a locally compact separable metric space. We define what it means for a function and a preorder to be subpeak continuous.

Subpeak continuous functions. A set M is a *locally satiated set* for $f : X \rightarrow \mathbb{R}$ if, for each $x \in \bar{M}$ there is some neighborhood N of x so that $f(x) \geq f(x')$ for all $x' \in N \cap M$. For an open set M we say that the function $f : X \rightarrow \mathbb{R}$ is *M -subpeak continuous* if f is continuous off the locally satiated set M . We say f is subpeak continuous if f is M -subpeak continuous for some open M . Subpeak continuity sits between regular continuity and upper semicontinuity in the following sense.

Lemma 1. For a function $f : X \rightarrow \mathbb{R}$,

$$f \text{ continuous} \implies f \text{ subpeak continuous} \implies f \text{ upper semicontinuous} \quad (1)$$

From Lemma 1 we see that every subpeak continuous function f is upper semicontinuous. Importantly, this guarantees that $\sup\{f(x) : x \in B\}$ is always attained by some x provided B is compact. We next define what it means for a preorder to be subpeak continuous.

Subpeak Continuous Preorders. Let $\geq$ be a preorder on X . We say that $\geq$ is *continuous* if $\{(x, x') \in X^2 : x \geq x'\}$ is closed. For a set $Y \subseteq X$ we say that $\geq$ is *continuous on Y* if, for all $\bar{x} \in X$ the sets $\{(x, x') \in Y^2 : x \geq x'\}$, $\{x \in Y : x \geq \bar{x}\}$, and $\{x \in Y : \bar{x} \geq x\}$ are closed. A set M is a *locally satiated set* for $\geq$ if, for each $x \in \bar{M}$ there is some neighborhood N of x so that $x \geq x'$ for all $x' \in N \cap M$. For some open set M the preorder $\geq$ is *M -subpeak continuous* if $\geq$ is continuous off the locally satiated set M . We say $\geq$ is *subpeak continuous* if $\geq$ is M -subpeak continuous for some open set M . The following equivalencies are easy to verify.

$$\geq \text{ continuous} \iff \geq \text{ continuous on } X \iff \geq \text{ is } \emptyset\text{-subpeak continuous}$$

We next explore the connection between subpeak continuous functions and preorders.

Subpeak continuous functions and preorders. We say that f is $\geq$ -*increasing* if $x \geq (>) x'$ implies $f(x) \geq (>) f(x')$. We say that f is $\geq$ -*weakly increasing* if $x \geq x'$ implies $f(x) \geq f(x')$. Of course, every $\geq$ -increasing function is weakly increasing but not vice-versa. The preorder $\geq$ is *generated* by f if $\geq$ satisfies $x \geq x'$ if and only if $f(x) \geq f(x')$. It is easy to verify that an M -subpeak continuous function generates an M -subpeak continuous preorder. The converse is also true in the following sense.

Lemma 2. *Let $\geq$ be an M -subpeak continuous preorder. There exists a $\geq$ -increasing and M -subpeak continuous function $f : X \rightarrow \mathbb{R}$.*

1.2 Utility and GARP

From hereon we assume that X is endowed with some subpeak continuous partial order $\succeq$. We say that $f : X \rightarrow \mathbb{R}$ is *increasing* if it is $\succeq$ -increasing and we say f is *weakly increasing* if it is $\succeq$ -weakly increasing. A function $U : X \rightarrow \mathbb{R}$ is called a *utility function*. A *dataset* is a collection $D = (x^t, B^t)_{t \in T}$ where T is an arbitrary index set, $x^t \in X$ is an alternative chosen from a non-empty compact choice set $B^t \subseteq X$. A utility function $U : X \rightarrow \mathbb{R}$ *rationalizes* D if

$$U(x^t) \geq U(x), \quad \forall t \in T \text{ and all } x \in B^t$$

Given two utility functions U and U' we say that U is *sharper* than U' if, for all t and all $x \in B^t$,

$$U'(x^t) > U'(x) \quad \implies \quad U(x^t) > U(x)$$

We say that x is *directly revealed preferred* to x' , denoted $x \succeq_D x'$, if either (i) $x \supseteq x'$ or (ii) $x = x^t$ and $x' \in B^t$ for some $t \in T$. The *revealed preference relation* $\succeq_D^*$ is the transitive closure of $\succeq_D$. We say that x is *strictly directly revealed preferred* to x' , denoted $x \succ_D x'$ if $x \succeq_D \bar{x} \supset x'$ for some $\bar{x}$. We say that D satisfies the *generalized axiom of revealed preferences* (GARP) if $x^t \succ_D x^s$ implies $x^s \not\succeq_D^* x^t$ for all $s, t \in T$.

It is easy to see that GARP is a necessary condition for D to be rationalized by an increasing utility function. Indeed, if U is increasing and rationalizes D then $x^t \succ_D x^s$ implies $U(x^t) \geq U(x^s)$ and $x^s \succeq_D^* x^t$ implies $U(x^t) \geq U(x^s)$ and so it cannot be the case that both $x^t \succ_D x^s$ and $x^s \succeq_D^* x^t$.

1.3 Coarse Utility

A function $V : \mathcal{A} \rightarrow \mathbb{R}$ is a *coarse utility function* if $\mathcal{A}$ is a partition of X . Every coarse utility function $V : \mathcal{A} \rightarrow \mathbb{R}$ generates a utility function U via

$$U(x) = V([x]) \tag{2}$$

where $[x]$ denotes the element of $\mathcal{A}$ to which x belongs. Additionally, if $U : X \rightarrow \mathbb{R}$ is constant on each $A \in \mathcal{A}$ then U generates the coarse utility function V also defined via (2). We say that V is *subpeak continuous* if the utility function it generates is subpeak continuous. We write $[x] \succ [x']$ in the event that $x \supset x'$ and $[x] \neq [x']$. We say $V : \mathcal{A} \rightarrow \mathbb{R}$ is *increasing* if $[x] \supset [x']$ implies $V([x]) > V([x'])$. A coarse utility function $V : \mathcal{A} \rightarrow \mathbb{R}$ *rationalizes* D if

$$V([x^t]) \geq V([x]), \quad \forall t \in T \text{ and all } x \in B^t$$

A coarse utility function $V : \mathcal{A} \rightarrow \mathbb{R}$ is *finer* than $V' : \mathcal{A}' \rightarrow \mathbb{R}$ if, for each $x \in X$, we have $[x]_{\mathcal{A}} \subseteq [x]_{\mathcal{A}'}$. Given two coarse utility functions V and V' we say that V is *sharper* than V' if, for all $t \in T$ and $x \in B^t$,

$$V'([x^t]) > V'([x]) \quad \implies \quad V([x^t]) > V([x])$$

Let us write $x \approx_D x'$ if either (i) $x \supseteq x'$ and $x' \succeq_D^* x$ or (ii) $x' \supseteq x$ and $x \succeq_D^* x'$. Let $\approx_D^*$ denote the transitive closure of $\approx_D$. Note that if an increasing coarse utility function $V : \mathcal{A} \rightarrow \mathbb{R}$ rationalizes D and $x \supset x'$ then either $V([x]) > V([x'])$ or $[x] = [x']$. Now,

if it is the case that $x \geq_D^* x'$ then we must have $V([x']) \geq V([x])$ and so we conclude that $x \sim_D x'$ implies $[x] = [x']$ and thus, we also see that $x \sim_D^* x'$ implies $[x] = [x']$. Let $\mathcal{A}^* = X / \approx_D^*$.

Theorem 1. *Suppose the revealed preference relation $\geq_D^*$ is M -subpeak continuous. Then, there is an increasing and M -subpeak continuous coarse utility function $V^* : \mathcal{A}^* \rightarrow \mathbb{R}$ which rationalizes D and V^* is finer and sharper than any other increasing coarse utility function which rationalizes D .*

Corollary 1. *Suppose $\geq_D^*$ is subpeak continuous. Then, D can be rationalized by an increasing subpeak utility function U if and only if D satisfies GARP.*

Proposition 1. *If D is finite then $\geq_D^*$ is subpeak continuous.*

Proof. A corollary of Lemma 3. □

For a dataset D let $\mathcal{V}_D$ be the collection of increasing coarse utility functions which rationalize D and let $\mathcal{V}_D^*$ be the collection of finest increasing subpeak continuous coarse utility functions which rationalize D . Here is a straightforward fact.

$$x \geq_D^* x' \implies V([x]) \geq V([x']), \quad \forall V \in \mathcal{V}_D \quad (3)$$

$$x >_D^* x' \implies V([x]) > V([x']), \quad \forall V \in \mathcal{V}_D^* \quad (4)$$

For sets $B, B' \subseteq X$ and binary relation $\geq$ let $B \geq B'$ mean that for all $x' \in B'$ there exists $x \in B$ so that $x \geq x'$.

Theorem 2. *Let D be a dataset and let B, B' be two non-empty compact subsets of X . Suppose $\geq_D^*$ is subpeak continuous. The following are equivalent.*

1. $\sup_{x \in B} V([x]) \geq \sup_{x' \in B'} V([x'])$ for all $V \in \mathcal{V}_D$.
2. $\sup_{x \in B} V([x]) \geq \sup_{x' \in B'} V([x'])$ for all $V \in \mathcal{V}_D^*$.
3. $B \geq_D^* B'$.

Also, the following are equivalent.

1. $\sup_{x \in B} V([x]) > \sup_{x' \in B'} V([x'])$ for all $V \in \mathcal{V}_D^*$.
2. $B >_D^* B'$.

For a non-empty B let $S(B)$ be the set

$$S(B) = \left\{ x \in B : V([x]) = \sup_{x' \in B} V([x']) \text{ for some } V \in \mathcal{V}_D^* \right\}$$

Corollary 2. *Let D be a dataset and let B be a non-empty compact subset of X . Suppose $\geq_D^*$ is subpeak continuous. Then, $x \in S(B)$ if and only if $B \not\prec_D^* x$.*

Proof. From the second part of Theorem 2 (just negate the two equivalent statements in the theorem) we see that $B \succ_D^* x$ if and only if $V([x]) \geq \sup_{x' \in B} V([x'])$ for some $V \in \mathcal{V}_D^*$ and so the corollary holds. $\square$

1.4 The Market

From now on we specialize to the classic case of L goods which can be purchased in any positive quantity and where $\succeq = \geq$. So, $X = \mathbb{R}_+^L$. A purchase dataset is a collection $D = (p^t, x^t)_{t \in T}$. Each purchase dataset D is identified with the dataset $(x^t, B^t)_{t \in T}$ where B^t is the linear budget set

$$B^t = \{x \in \mathbb{R}_+^L : p^t \cdot x^t \geq p^t \cdot x\}$$

The compensating variation of a price change. Let $CV_V^t(\bar{p})$ be the additional money (over $p^t \cdot x^t$) needed to get a consumer with coarse utility V the same utility as $V([x^t])$. That is,

$$CV_V^t(\bar{p}) = \inf \{\bar{p} \cdot x : V([x]) \geq V([x^t])\} - p^t \cdot x^t$$

Also, let

$$CV_D^t(\bar{p}) = \inf \{\bar{p} \cdot x : x \succeq_D^* x^t\} - p^t \cdot x^t$$

Similarly, the strict compensating variation $SV_V^t(\bar{p})$ is the additional money needed to get a consumer with coarse utility V strictly more utility than $V([x^t])$. That is,

$$SV_V^t(\bar{p}) = \inf \{\bar{p} \cdot x : V([x]) > V([x^t])\} - p^t \cdot x^t$$

Also, let

$$SV_D^t(\bar{p}) = \inf \{\bar{p} \cdot x : x \succ_D^* x^t\} - p^t \cdot x^t$$

Proposition 2. Suppose $D = (p^t, x^t)_{t \in T}$ is a purchase dataset so that $\succeq_D^*$ is subpeak continuous. For $t \in T$ and $\bar{p} \in \mathbb{R}_{++}^L$

$$\sup_{V \in \mathcal{V}_D} CV_V^t(\bar{p}) = \sup_{V \in \mathcal{V}_D^*} CV_V^t(\bar{p}) = CV_D^t(\bar{p}) \quad (5)$$

and

$$\sup_{V \in \mathcal{V}_D^*} SV_V^t(\bar{p}) = SV_D^t(\bar{p}) \quad (6)$$

2 Proofs

2.1 Lemmas

Proof of Lemma 1. If f is continuous then f is clearly $\emptyset$ -subpeak continuous and thus subpeak continuous. Now, suppose f is M -subpeak continuous for some open M . Let x_n be a sequence converging to some x . If x_n takes infinitely many values outside of M then by definition of subpeak continuity we have $\lim f(x_n) = f(x)$. If, on the other hand, x_n takes all but finitely values in M then, for some N , we have $f(x) \geq f(x_n)$ for all $n \geq N$ and so $f(x) \geq \limsup f(x_n)$ and so f is upper semicontinuous. $\square$

Proof of Lemma 2. Let $\sim$ be the symmetric part of $\geq$. Let $\mathcal{M} = M / \sim$ and endow $\mathcal{M}$ with the discrete topology. Let $Y = X \setminus M$ and let $\tilde{X} = Y \sqcup \mathcal{M}$ (that is, $\tilde{X}$ is the disjoint union of Y and $\mathcal{M}$). Let $\phi : X \rightarrow \tilde{X}$ satisfy $\phi(x) = x$ if $x \in Y$ and $x \in \phi(x)$ if $x \in M$. Also, let $\psi : \tilde{X} \rightarrow X$ satisfy $\psi(x) = x$ if $x \in Y$ and $\psi(x) \in x$ if $x \in \mathcal{M}$. Let $\tilde{\geq}$ be the preorder defined on $\tilde{X}$ by $\tilde{x} \tilde{\geq} \tilde{x}'$ if and only if $\psi(\tilde{x}) \geq \psi(\tilde{x}')$.

We show that $\mathcal{M}$ has countably many members. Note that X is second countable and so it has a countable base $\mathcal{E}$. Now, for every $m \in \mathcal{M}$ there is some open set $E_m \in \mathcal{E}$ so that $\psi(m) \in E_m$ and $\psi(m) \geq x$ for all $x \in E_m$. Clearly, if $m \neq m'$ then $E_m \neq E_{m'}$. Thus, there is a map from $\mathcal{E}$ onto $\mathcal{M}$ and so $\mathcal{M}$ is countable.

It is easy to see that $\tilde{X}$ is a locally compact separable metrizable space and that $\tilde{\geq}$ is a continuous preorder and so, by Levin's Theorem, there is some $\tilde{\geq}$ -increasing function and continuous $\tilde{f} : \tilde{X} \rightarrow \mathbb{R}$. Define $f : X \rightarrow \mathbb{R}$ by $f(x) = \tilde{f}(\phi(x))$. It is easy to see that f is $\geq$ -increasing, continuous on Y , and that M is a locally satiated set for f and so the proof is complete. $\square$

Lemma 3. *Let $D = (x^t, B^t)_{t \in T}$ and $\tilde{D} = (x^t, B^t)_{t \in \tilde{T}}$ be datasets where $T \subseteq \tilde{T}$ and $\tilde{T} \setminus T$ is finite. If $\geq_D^*$ is M -subpeak continuous then $\geq_{\tilde{D}}^*$ is also M -subpeak continuous.*

Proof. It clearly suffices to prove the special case where $\tilde{T} \setminus T$ contains a single element. So, suppose that $\tilde{T} \setminus T = \{\tilde{t}\}$. Let x_n and y_n be convergent sequences in $X \setminus M$ so that $x_n \geq_D^* y_n$. We must show that $\lim x_n \geq_{\tilde{D}}^* \lim y_n$. The result is clear if $x_n \geq_D^* y_n$ for infinitely many n (as $\geq_D^*$ is M -subpeak continuous) so let us assume that $x_n \not\geq_D^* y_n$ for all n . Thus, for each n there exists $\tilde{x}_n \in B^{\tilde{t}}$ so that $x_n \geq_D^* \tilde{x}_n$ and $\tilde{x}_n \geq_D^* y_n$. As $x_n \notin M$ we have $\lim x_n \geq_D^* \tilde{x}$. As $B^{\tilde{t}}$ is compact we may assume that $\tilde{x} = \lim \tilde{x}_n$ for some $\tilde{x} \in B^{\tilde{t}}$. If $\tilde{x}_n \notin M$ for infinitely many n then, as $\geq_D^*$ is M -subpeak continuous, we have $\tilde{x} \geq_D^* \lim y_n$ and so $\lim x_n \geq_D^* \lim y_n$. On the other hand, if $\tilde{x}_n \in M$ for all n then there must be some N so that $\tilde{x} \geq_D^* \tilde{x}_n$ for all $n \geq N$. Thus, $\tilde{x} \geq_D^* \lim y_n$ and so again $\lim x_n \geq_D^* \lim y_n$. $\square$

Lemma 4. Let $D = (x^t, B^t)_{t \in T}$ be a dataset, let $\geq_D^*$ be subpeak continuous, let $\tilde{x} \in X$, and $B \subseteq X$ be a compact set.

1. If $B \not\prec_D^* \tilde{x}$ then there exists a $V \in \mathcal{V}_D^*$ satisfying $V([\tilde{x}]) \geq \sup_{x \in B} V([x])$.
2. If $B \not\preceq_D^* \tilde{x}$ then there exists a $V \in \mathcal{V}_D^*$ satisfying $V([\tilde{x}]) > \sup_{x \in B} V([x])$.

Proof. Let D' be the dataset formed by appending the observation $(\tilde{x}, B \cup \{\tilde{x}\})$ to D . By Lemma 3 the preorder $\geq_{D'}^*$ is subpeak continuous.

Claim 1. If $B \not\prec_D^* \tilde{x}$ then $\sim_D^* = \sim_{D'}^*$.

Proof of Claim 1. As every observation in D' is also contained in D we have $\geq_D^* \subseteq \geq_{D'}^*$. Thus, $\sim_D^* \subseteq \sim_{D'}^*$. So, for a contradiction suppose that $\sim_D^*$ is a strict subset of $\sim_{D'}^*$. In other words, there is some $x \sim_{D'} x'$ so that $x \not\preceq_D^* x'$. This means that $x \geq_D^* \tilde{x} \geq_{D'} x'$. Thus, for some $\bar{x} \in B$ we have $\bar{x} \geq_D^* x'$. As $B \not\prec_D^* \tilde{x}$ we see that

$$x' \not\prec_D^* \tilde{x} \tag{7}$$

from which we see that $x' \not\prec_D^* x$ and thus $x' \not\preceq_D^* x$. As $x \sim_{D'} x'$ and $x' \not\preceq_D^* x$ we see that $x' \geq_D^* \tilde{x}$ but from (7) we see that $x \geq_D^* \tilde{x} \sim_{D'}^* x'$ which contradicts our assumption that $x \not\preceq_D^* x'$ and so $x \geq_D^* x'$. By symmetry we have $x' \geq_D^* x$ and so the proof is complete. $\square$

Suppose $B \not\prec_D^* \tilde{x}$. By Lemma 1 there is some $V \in \mathcal{V}_{D'}^*$ and so $V([\tilde{x}]) \geq \sup_{x \in B} V([x])$. From Claim 1 we know that $\sim_D^* = \sim_{D'}^*$ and so $V \in \mathcal{V}_D^*$.

Suppose further that $B \not\preceq_D^* \tilde{x}$. From Lemma 1 we know that V is upper semicontinuous and so there is some $\bar{x} \in B$ satisfying $V([\bar{x}]) = \sup_{x \in B} V([x])$. Because $V \in \mathcal{V}_D^*$ and $V \in \mathcal{V}_{D'}^*$ we see that $V([\tilde{x}]) > V([\bar{x}])$ which completes the proof. $\square$

2.2 Theorems and Propositions

Proof of Theorem 1. As $\geq_D^*$ is M -subpeak continuous we may appeal to Lemma 2 to find some $\geq_D^*$ -increasing function $U : X \rightarrow \mathbb{R}$. Note that U is constant on each element of $\mathcal{A}^*$ and so we can define $V^* : \mathcal{A}^* \rightarrow \mathbb{R}$ via (2). It is evident that V^* is M -subpeak continuous and so we must show that V^* is increasing, rationalizes D , and is sharper and finer than any other increasing V which also rationalizes D .

For $x \in X$ let $[x]$ denote the set in $\mathcal{A}^*$ to which x belongs. Because U is $\geq_D^*$ -increasing it is clear that V rationalizes D . Also, note that $[x] \triangleright [x']$ implies $x >_D^* x'$ and so $U(x) > U(x')$ and so $V([x]) > U([x'])$. Thus, V is increasing.

Let $V : \mathcal{A} \rightarrow \mathbb{R}$ be some increasing coarse utility function which rationalizes D . We shall show that V^* is finer and sharper than V . If $x \triangleright x'$ then either $V([x]) > V([x'])$ or

$[x] = [x']$. It follows that $x \approx_D x'$ implies $[x] = [x']$ and thus $x \approx_D^* x'$ implies $[x] = [x']$. From this it is clear that V^* is finer than V . Suppose $x \in B^t$ and $V([x^t]) > V([x])$. It must be that $x \not\geq_D^* x^t$. Therefore, $x^t >_D^* x$ and thus $V^*([x^t]) > V^*([x])$ and so V^* is sharper than V . $\square$

Proof of Theorem 2. We prove the first part of theorem. That 3 implies 1 follows from (3) and 1 implies 2 is obvious. So, let us assume that $B \not\geq_D^* B'$. We shall find some $V \in \mathcal{V}_D^*$ for which $\sup_{x' \in B'} V([x']) > \sup_{x \in B} V([x])$. As $B \not\geq_D^* B'$ there is some $x' \in B'$ so that $B \not\geq_D^* x'$. Lemma 4 provides the desired $V \in \mathcal{V}_D^*$.

Now we move on to the second part. Suppose $B >_D^* B'$ and let $V \in \mathcal{V}_D^*$. From Lemma 1 there is some $\tilde{x}' \in B'$ so that $V([\tilde{x}']) = \sup_{x' \in B'} V([x'])$. As $B >_D^* B'$ there exists some $\tilde{x} \in B$ so that $\tilde{x} >_D^* \tilde{x}'$. So, by (4) we see that $\sup_{x \in B} V([x]) \geq V([\tilde{x}]) > V([\tilde{x}'])$.

Next, suppose that $B \not\geq_D^* B'$. This means that, for some $x' \in B'$, we have $B \not\geq_D^* x'$. Lemma 4 then establishes that there is some $V \in \mathcal{V}_D^*$ so that $V([x']) \geq \sup_{x \in B} V([x])$. $\square$

Proof of Proposition 2. From (3) we see that $CV_D^t(\bar{p}) \geq CV_V^t(\bar{p})$ for all $V \in \mathcal{V}_D$. Thus, $CV_D^t(\bar{p}) \geq \sup_{V \in \mathcal{V}_D} CV_V^t(\bar{p}) \geq \sup_{V \in \mathcal{V}_D^*} CV_V^t(\bar{p})$. So, we must show that $\sup_{V \in \mathcal{V}_D^*} CV_V^t(\bar{p}) = CV_D^t(\bar{p})$.

Let $m = \inf_{x \in \mathbb{R}_+^L} \{p \cdot x : x \geq_D x^t\}$, let $\varepsilon > 0$, and let $B = \{x \in X : m - \varepsilon \geq \bar{p} \cdot x\}$. We have $B \not\geq_D^* x^t$ and so Lemma 4 provides some $V \in \mathcal{V}_D^*$ which satisfies $V([x^t]) > \sup_{x \in B} V([x])$. Therefore, $CV_V^t(\bar{p}) \geq CV_D^t(\bar{p}) - \varepsilon$ and because $\varepsilon > 0$ was arbitrary we conclude that $\sup_{V \in \mathcal{V}_D^*} CV_V^t(\bar{p}) = CV_D^t(\bar{p})$.

From (4) we see that $SV_D^t(\bar{p}) \geq SV_V^t(\bar{p})$ for all $V \in \mathcal{V}_D^*$ and so $SV_D^t(\bar{p}) \geq \sup_{V \in \mathcal{V}_D^*} SV_V^t(\bar{p})$. Let $m = \inf_{x \in \mathbb{R}_+^L} \{p \cdot x : x >_D x^t\}$, $\varepsilon > 0$, and $B = \{x \in X : m - \varepsilon \geq \bar{p} \cdot x\}$. Proceeding as before we can find some $V \in \mathcal{V}_D^*$ which satisfies $SV_V^t(\bar{p}) \geq SV_D^t(\bar{p}) - \varepsilon$ and so (6) holds. $\square$

References